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An attention mechanism and residual network based knowledge graph-enhanced recommender system

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ABSTRACT

Recommender systems enhanced by a knowledge graph (KG) have attained widespread popularity and attention in recent years. However, traditional KG-based recommender systems encounter the challenge of gradient explosion as the network depth increases. Additionally, the abundance of unreliable paths in a KG has a detrimental impact on feature representation learning. In this article, we propose a KG-enhanced recommender system based on residual network and attention mechanism, which can capture high-order connectivity and long-range dependencies of the KG. Specifically, a resource allocation approach is employed to calculate the resource amount, which is subsequently utilized to evaluate the path reliability of the KG. After completing path extraction, we employ an attention mechanism to capture semantic correlations and structural information. To leverage the KG for enhancing recommender systems, we design a deep residual network with shortcut connections, effectively amalgamating advanced and abstract features using deep neural networks. The introduction of shortcut connections not only facilitates the fitting of residual mappings but also mitigates potential issues such as gradient explosion and convergence difficulties due to excessive network depth. Extensive experiments conducted on three standard datasets over baseline methods have demonstrated the superiority of our proposed recommender system.

1. Introduction

Recommender systems [1,2] have become an indispensable part of modern online platforms in the era of information overload. In recent years, the introduction of a knowledge graph (KG) to enhance recommender systems has gained significant popularity. KG [3,4] is a heterogeneous information network, encompassing a wealth of entity, relations, and side information, assisting recommender systems in accurately capturing the relationships and preferences between users and items. The KG-enhanced National Basketball Association (NBA) stars recommender system is taken as an example in Fig. 1. The scenario where Tom holds an affinity for Michael Jordan and Kobe Bryant is considered. The external KG shows that Tom may be interested in NBA stars Dwyane Wade and Shaquille O'Neal.

To utilize KG information to enhance recommendations, researchers have explored many KG-based recommender systems [5]. The early methods primarily focused on mining implicit interactions among entities through the paths in knowledge graphs (KGs). PathSim [6] represents specific paths connecting two entities as meta-paths [7] and employs path similarity methods to mine graph information. FMG [8] extends the concept of meta-paths by employing more informative meta-graphs to capture potential correlations between entities. As a follow-up, translation-based methods employ score functions to evaluate the plausibility of facts between two entities. For example, TransE [9] executes the formula $h + r \approx t$ into the embedding space, with h, r, and t representing the head entity, relation, and tail entity, respectively. Many other translation-based methods like TransH [10] and TransR [11] expand upon TransE and tackle its limitations when dealing with 1-to-N, N-to-1, and N-to-N relations.

While translation-based methods offer effective and intuitive strategies, they often encounter performance degradation when tasked with modeling different relations involving the same entities in complex scenarios. The advent of matrix factorization has led to numerous advancements in the field of graph representation learning. FM [12] utilizes matrix factorization to model high-dimensional sparse data, allowing it to efficiently model complex relationships between features. NFM [13] is an extension of FM that incorporates neural networks to handle sparse data and capture high-order feature interactions. Methods based on semantic matching [14–16] are also widely favored. For instance, RESCAL [17] adopts a three-way tensor decomposition

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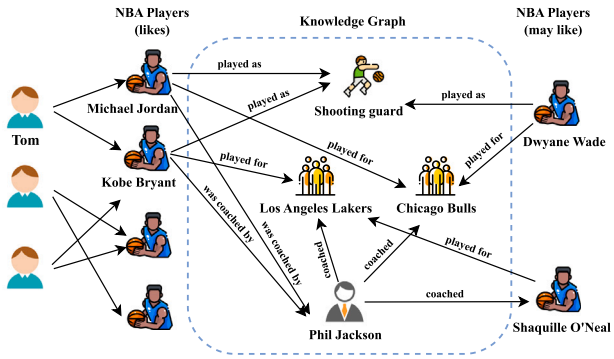


Fig. 1. An example of KG enhanced NBA stars recommender system. Given an NBA basketball KG that encompasses entities such as player, team, coach, position, and their interrelationships. Due to both Michael Jordan and Dwyane Wade playing as shooting guards and have played for the Chicago Bulls, we can assume that Tom may like Dwyane Wade. Similarly, both Kobe Bryant and Shaquille O'Neal have played for Los Angeles Lakers and have been coached by Phil Jackson, we can recommend to Tom the player Shaquille O'Neal.

to model a graph structure, enabling collective learning through latent components of the model. DistMult [18] leverages the learned representations to mine logical rules for multi-relational learning. HOLE [19] utilizes the holographic embedding approach to search for semantic connections among entities and relations. ComplEx [20] effectively handles a wide range of binary relations using the Hermitian dot product. InteractE [21] leverages recurrent convolution to enhance the number of interactions. However, the abovementioned methods still struggle to capture higher-order semantics and handle sparse graph data, and they often rely on specific graph structures or external semantic resources.

In recent years, neural network based methods have drawn increasing interest and popularity in various graph analytics tasks. KGAT [22] employs attention networks to dynamically weigh the importance of different relations between a user and items. ConvE [23] utilizes convolutional neural networks (CNNs) [24] to capture complex patterns and interactions between entities. RippleNet [25] integrates the KG into recommender systems by propagating user preferences along links, forming preference distribution for predicting item clicks. KGCN [26] samples multi-hop neighbors of the candidate items from the KG and conducts a fixed number of iterations from each entity. ResNets [27] adopt residual networks with skip connections to alleviate the vanishing gradient problem in deep neural networks. MCRec [28] utilizes meta-path based contextual information and the neural co-attention mechanism for affording the top-N recommendation. RKGE [29] automatically mines semantic information of entities and path features to characterize user preferences toward items. EIUM [30] leverages semantic paths of a KG to provide path interpretation and captures user preferences at the interaction level through multimodal fusion techniques. Despite the successes achieved by neural network based methods, most existing approaches encounter problems with gradient explosion and network convergence when attempting to deepen the networks. As the number of network layers increases, the capacity to capture abstract and advanced features also increases. However, the deeper the neural network, the more challenging the optimization. Additionally, the presence of numerous unreliable paths in a KG can negatively impact the robustness and accuracy of these models.

In this article, we propose an attention mechanism and residual network (AMRN) based KG-enhanced recommender system. First, the relation resource amount is calculated to evaluate the path reliability in the KG. It not only provides higher-quality structural information for subsequent graph representation learning but also alleviates the sparsity of the KG. Second, an attention mechanism is applied to learn representations for users and items, allowing for capturing higher-order information and long-range dependencies of the KG. Finally, we employ a residual network with skip connections, which consists of a mainstem and shortcut connections. The mainstem utilizes deep CNNs to learn comprehensive information, while the shortcut connections mitigate the issues of gradient explosion and vanishing gradients by allowing skips to subsequent network layers.

The main contributions of this paper can be summarized as follows:

- We develop a resource allocation algorithm to assess path reliability and efficiently filter out unreliable paths within the KG. An attention mechanism is further applied to the KG following path extraction, facilitating the capture of higher-order connectivity and long-range dependencies.
- We propose a KG-enhanced recommender system via a deep residual network based on shortcut connections, which not only deeply exploits the feature information of the KG but also mitigates the problem of gradient explosion, gradient vanishing, and convergence difficulties.
- Comprehensive experiments are conducted on three real-world datasets, and the results demonstrate that the AMRN achieves efficient performance over existing baselines.

The rest of this paper is organized as follows. Section 2 gives a brief overview of related work. Section 3 illustrates the details of our proposed method AMRN. Extensive experiments are reported in Section 4. Finally, we summarize the conclusions of this paper in Section 5.

2. Related work

In this section, KG-based recommendation methods are mainly classified into three categories: path-based methods, embedding-based methods, and unified methods.

2.1. Path-based methods

Path-based methods utilize user-item interactions and extract hidden associations within a KG to perform recommendations. A meta-path refers to a predefined sequence of entity types and relation types that describe a specific type of path or connectivity pattern. Meta-paths are used to capture and formalize the semantic structure of relations between entities in KG. PathSim [6] leverages meta-path based similarity measures to capture intricate relationships between users and items, enhancing the accuracy of recommendations. HeteRec [31] utilizes a meta-path based random walk and extended matrix factorization model to integrate various information embedded in a heterogeneous information network to improve recommendation performance. FMG [8] introduces the concept of meta-graphs to effectively capture latent features of various meta-graphs through matrix factorization. SemRec [32] employs weighted meta-paths to enhance the representation of path semantics for recommender systems. RippleNet [25] is an end-to-end framework that iteratively extends the latent interests of users by connecting along paths in a KG, forming a preference distribution for candidate items based on historical interactions.

2.2. Embedding-based methods

The purpose of KG embedding [33–35] is to employ mapping functions that convert relations and entities into continuous and low-dimensional vector representations. CKE [36] is a KG-based embedding model within a collaborative filtering framework, CKE utilizes TransR [11] to embed the latent vectors of structural knowledge associated with items, followed by feature extraction using autoencoders. FM [12] employs matrix factorization to efficiently model high-dimensional sparse data while dealing with interactions between features. NFM [13] builds upon FM by incorporating neural networks to mine for higher-order feature interactions, addressing the issue

of sparse feature interactions in recommendation tasks. CFKG [37] combines knowledge base embedding and soft matching algorithms to generate personalized explainable recommendations. RuleRec [38] leverages item-centric KGs to provide human-readable explanations and improve generalization for cold-start scenarios. KPRN [39] designs path-based reasoning to exploit sequential dependencies and user-item interactions.

2.3. Unified methods

Embedding-based methods employ semantic vectors of users and items from KG for recommendation, while path-based methods focus on utilizing path information via meta-paths to reveal deeper associations. Researchers aim to integrate these methods to enhance entity representation and effectively manage large-scale KGs, addressing challenges in meta-path selection, vector learning, and efficient operations. KGAT [22] models higher-order relationships between users and items by leveraging TransR [11] to obtain initial entity embedding and iteratively propagating information from entities to multi-hop neighbors. KR-GCN [40] utilizes a transition-based approach to determine the tertiary scores and uses kernel sampling to adaptively select the triples within the path between each user-item pair. JNSKR [41] employs a non-sampling optimization strategy for learning graph representations. It utilizes a knowledge-aware CNN to encode subgraphs, effectively describing user preferences for items. To address the issues of hash collisions and embedding sparsity, CERP [42] combines two meta-embedding tables and performs joint pruning with regularization to encode mutually complementary information and achieve accurate recommendations. CGAT [43] uses local and non-local graph contextual information in KGs for enhancing personalized recommendations. PSA-GNN [44] incorporates the pre-mined subgraphs as priori knowledge to build a hybrid receptive field of GNNs, sufficiently exploiting the higher-order subgraph information.

Despite their effectiveness, the existing methods in each category have their limitations. Path-based methods may struggle with scalability when applied to large and complex KGs, as exploring all possible paths can become computationally expensive. Meanwhile, most embedding-based methods exhibit limitations in effectively capturing higher-order information in KGs, and when attempting to deepen the network for advanced feature extraction, they often encounter issues like gradient explosion, gradient vanishing, or convergence difficulties. To tackle these challenges, we propose a unified model AMRN, which leverages the reliability of paths for path extraction and utilizes attention mechanisms along with a shortcut-based residual network to jointly achieve representation learning. AMRN aims to handle the limitations of traditional unified methods by effectively capturing high-order information in KGs while mitigating issues related to network depth and convergence.

3. The proposed model

In this section, we present a thorough explanation of our proposed AMRN, including preliminaries, an overview of AMRN, attention mechanism, residual network based on shortcut connections, and training AMRN.

3.1. Preliminaries

A recommender system is designed to analyze user preferences and behaviors to provide personalized recommendations of items for users. The abundant interaction information of items in the real world often forms a KG, providing richer contextual information for recommender systems. The KG can be constructed from the side information of items, such as attributes and external knowledge. These auxiliary data comprise entities and their relations associated with items, thereby enhancing the performance of the recommender systems. Graph representation embedding is further utilized to transform entities and relations into low-dimensional vector spaces, capturing the complex semantic information of user-item interactions. For clarity, the main notations used in this article are summarized in Table 1.

Table 1
Summary of the main notations.

Notations	Description
U, I	The set of users and items
R	The matrix of user-item interaction
$F : U \times I \to \mathbb{R}$	The estimation function for user, item
$(h, r, t) \in \mathcal{T}$	The triple of head, relation, and tail
$\mathcal{E}$, $\mathcal{R}$, $\mathcal{T}$	The set of entities, relations and triples
$\mathcal{G}_{KG} = (\mathcal{E}, \mathcal{R}, \mathcal{T})$	A knowledge graph
$\mathcal{G}_{UI}$	A user-item interaction graph
$\phi_e : \mathcal{E} \to \mathbb{R}^d$	The mapping function for entities
$\phi_r : \mathcal{R} \to \mathbb{R}^d$	The mapping function for relations
T^-	The set of invalid triplets
$p = (r_1, \dots, r_n)$	The relation path
b	The number of residual blocks
s	The reliability score
d	The embedding dimension

Definition 1 (*Recommender System*). The historical interactions between users and items can be captured in a user-item interaction matrix R, where R_{ui} denotes the interaction of user u with item i. The objective of a recommender system is to execute the function $F : U \times I \to \mathbb{R}$, which aims to estimate the missing or unobserved entries in R, ie., $F(u, i)$ estimates the affinity of user u for item i.

Definition 2 (*Knowledge Graph*). A Knowledge Graph can be defined as a directed graph $\mathcal{G}_{KG} = (\mathcal{E}, \mathcal{R}, \mathcal{T})$, where $\mathcal{E}$ represents entities, $\mathcal{R}$ represents relations between entities, and $\mathcal{T}$ is the set of triples (h, r, t), where h and t are entities from $\mathcal{E}$, and r from $\mathcal{R}$ represents a directed relation from h to t.

Definition 3 (*Graph Representation Learning*). Graph representation learning for KGs employs mapping functions $\phi_e : \mathcal{E} \to \mathbb{R}^d$ and $\phi_r : \mathcal{R} \to \mathbb{R}^d$ to transform entities and relations into d-dimensional and continuous vector spaces, respectively. The learned vectors $\phi_e(h)$, $\phi_r(r)$, and $\phi_e(t)$ corresponding to the triple $(h, r, t) \in \mathcal{T}$ reflect geometric relationships and semantics in the vector space.

3.2. Overview of the AMRN

The overview of our proposed model AMRN is depicted in Fig. 2. The first step involves developing a resource allocation method, which aims at evaluating the reliability of relation paths in KGs. Unreliable paths can be identified and filtered through the resource propagation of the paths. After completing the path extraction, an attention mechanism is adopted to capture higher-order connectivity and long-range dependencies in the KG, contributing to more comprehensive representations. On the other hand, KGs in collaboration with items are utilized to enhance item embedding for improving recommendation performance. We a design deep residual network with shortcut connections to extract structural features and semantic information of KGs, and also alleviate the problem of gradient vanishing and gradient exploding. By concatenating item embedding of KG-enhanced representations and user-item interactions, more representative item embedding can be obtained. The concatenated item embedding is then utilized to compute the inner product with the user embedding for recommendation. Subsequently, the concatenated item embedding undergoes a feature fusion and extraction process via a fully connected layer. The resulting item embedding is then combined with the user representation through an inner product operation for recommendations.

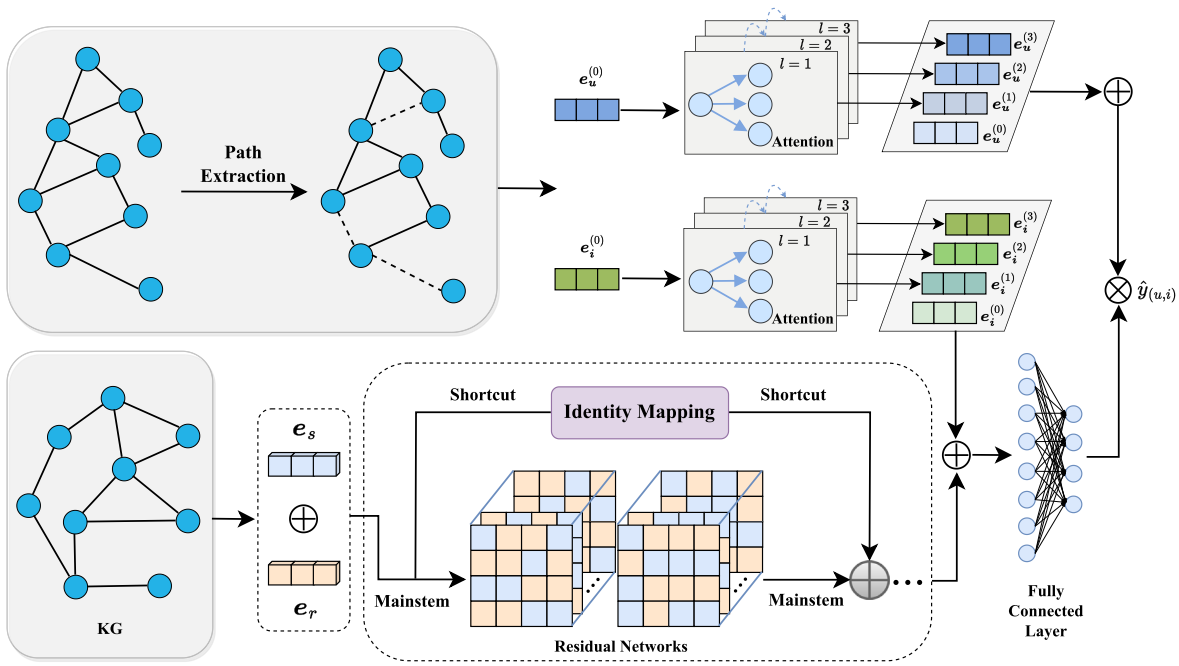


Fig. 2. The framework of AMRN. Note here that AMRN is composed of path extraction, an attention mechanism, and deep residual networks with shortcut connections. First, a resource allocation algorithm is employed to evaluate the path reliability of KG for path extraction. Then, we utilize an attention mechanism to capture semantic correlations and structural information of the KG. A deep residual network with shortcut connections is further utilized to effectively fuse advanced and abstract features. Finally, the embedding of items is concatenated and then subjected to an inner product operation with the embedding of users for recommendation.

3.3. Path extraction

The graph structure contains numerous paths, i.e., edges, which contribute considerable interconnected information that is crucial for graph representation learning. However, not all paths can be deemed reliable, with some of them even introducing unwanted noise. The motivation behind measuring path reliability lies in the demand to implement path extraction by filtering out unreliable paths in KGs. Inspired by literature [45], path reliability can be calculated through a resource allocation approach. Given a path triple (h, p, t), the amount of resource propagation from entity h to entity t could be calculated by multiple-step relation paths $(r_1, \dots, r_n)$. The resource propagation of the path is denoted as $\lambda_0 \xrightarrow{r_1} \lambda_1 \xrightarrow{r_2} \cdots \xrightarrow{r_n} \lambda_n$, with λ_0 representing the initial entity h and t being part of λ_n. Therefore, the amount of resource propagation obtained from the entity $i \in S_m$ can be calculated as:

$$R_p(i) = \sum_{j \in S_{m-1}(\cdot,i)} \frac{1}{|S_m(j,\cdot)|} R_p(j), \tag{1}$$

where $S_{m-1}(\cdot, i)$ is the direct predecessors in S_{m-1} along the relation r_m, and $S_m(j, \cdot)$ denotes the direct successors of $n \in S_{m-1}$ following the relation r_m. The amount of resource propagation $R_p(t)$ indicates how much information the head entity h can transmit to the tail entity t, serving as a measure for the reliability of the path p from h to t.

Then, a set of invalid triplets T^- can be obtained as described below:

$$T^- = \{(h', r, t)\} \cup \{(h, r', t)\} \cup \{(h, r, t')\}. \tag{2}$$

Ultimately, incorporating invalid triplet sets completes the path extraction process by refining the structure of KG, improving the overall integrity and utility of the extracted paths.

3.4. Attention mechanism

To assess the significance or relevance of edges (or other elements) within a given context, we employ an attention mechanism to obtain an attentive weight associated with each neighbor in an embedding propagation process. Specifically, the attention score of triplets according to the semantic distance is calculated as:

$$\delta(h, r, t) = (\mathbf{W}_r \mathbf{e}_t)^\top \tanh\left((\mathbf{W}_r \mathbf{e}_h + \mathbf{e}_r)\right), \tag{3}$$

where $\mathbf{W}_r \in \mathbb{R}^{k \times d}$ is the transformation matrix of relation r, $\tanh(\cdot)$ is a nonlinear activation function, $\mathbf{e}_t$, $\mathbf{e}_t \in \mathbb{R}^d$ and $\mathbf{e}_r \in \mathbb{R}^k$ are the embedding for h, t, and r, respectively.

The attention score of the current triplet is determined by the distance between the head and tail entities in the relational vector space, indicating that more similar entities receive more information. The softmax function is then used to normalize the computed results for each triplet linked to h:

$$\delta(h, r, t) = \frac{\exp(\delta(h, r, t))}{\sum_{(h,r',t') \in \mathcal{N}_h} \exp\left(\delta\left(h, r', t'\right)\right)}. \tag{4}$$

Let $\mathcal{N}_h = \{(h, r, t) \mid (h, r, t) \in \mathcal{G}_{KG}\}$ denotes the set of triples with the head entity h, the embedded representation of the set of triplets around h is calculated as:

$$\mathbf{e}_{\mathcal{N}_h} = \sum_{(h,r,t) \in \mathcal{N}(h)} \delta(h, r, t) \mathbf{e}_t. \tag{5}$$

To capture the first-order connectivity of the KG, we apply GCN to aggregate the entity representation $\mathbf{e}_h$ and neighborhood representations $\mathbf{e}_{\mathcal{N}_h}$ into a new representation of h, which is formulated as follows:

$$\begin{aligned} \mathbf{e}_h &= \mathrm{AGG}(\mathbf{e}_h, \mathbf{e}_{\mathcal{N}_h}) \\ &= \mathrm{LeakyReLU}\left(\mathbf{W}\left(\mathbf{e}_h + \mathbf{e}_{\mathcal{N}_h}\right) + \mathbf{b}\right), \end{aligned} \tag{6}$$

where $\mathbf{W}$ is the learnable parameter matrix in the nonlinear transformation, $\mathrm{LeakyReLU}(\cdot)$ is the activation function, and $\mathbf{b}$ is the bias vector.

Subsequently, we stack more propagation layers to exploit higher-order connectivity information and model semantic information from high-hop neighborhoods. The representation of entities in the lth order neighborhood of the head entity h can be represented as:

$$\mathbf{e}_h^{(l)} = f\left(\mathbf{e}_h^{(l-1)}, \mathbf{e}_{\mathcal{N}_h}^{(l-1)}\right). \tag{7}$$

After performing l layers of information propagation and aggregation for the user entity u based on the above process, multiple representations of u are obtained, i.e., $\{\mathbf{e}_u^1, \dots, \mathbf{e}_u^l\}$. As the output of layer l results from information aggregation on the l-layer tree structure rooted at user u (or item i), the outputs of different layers highlight connectivity information in varying orders. This not only expands the higher-order information from the initial embedding but also controls the depth of higher-order propagation by adjusting l. To maximize information preservation, we concatenate the representations obtained from each layer into a single vector, the formula is as follows:

$$\mathbf{e}_u^* = \mathbf{e}_u^1 \parallel \cdots \mathbf{e}_u^{(l-1)} \parallel \mathbf{e}_u^l, \quad \mathbf{e}_i^* = \mathbf{e}_i^1 \parallel \cdots \mathbf{e}_i^{(l-1)} \parallel \mathbf{e}_i^l, \tag{8}$$

where $\parallel$ represents concatenation operation. Specifically, $\mathbf{e}_u^*$ represents the concatenation of l layers of user embedding. Similarly, $\mathbf{e}_i^*$ represents the concatenation of l layers of item embedding. Additionally, the item representation is further concatenated with the enhanced representation from KGs.

3.5. Residual network based on shortcut connections

KG is a vast repository of structured information, comprising a multitude of meaningful information in a semantic structure. To incorporate knowledge regarding entities and relations in KGs into item representations, we design a deep residual network with shortcut connections to effectively capture structural and semantic information of KGs for enhancing item representation. As shown in Fig. 2, the deep residual networks based on shortcut connections consist of two main components: mainstem and shortcut connections.

During the forward propagation, the embedding of entities and relations are concatenated and then fed into the mainstem and the shortcut connections. In contrast to models that employ a single-layer 2D CNN, the mainstem of the residual network in AMRN utilizes a two-layer CNN to increase the network's depth. This enhances the capacity of representation learning and generalization. The residual mapping function can be denoted as:

$$\mathcal{F}(\boldsymbol{x}) = \mathbf{W}_2(\mathbf{W}_1\boldsymbol{x} + \mathbf{b}_1) + \mathbf{b}_2, \tag{9}$$

where $\boldsymbol{x}$ is the input embedding, $\mathbf{W}_1$ and $\mathbf{W}_2$ denote the weight matrix, $\mathbf{b}_1$ and $\mathbf{b}_2$ denote the bias parameter.

As is well-known, increasing the depth of networks enables more abstract combinations of high-level features. However, deeper networks are susceptible to issues like gradient explosion or gradient vanishing, which can be challenging to converge. To mitigate these problems, the introduction of shortcut connections helps fit residual mappings, effectively performing identity mapping. Linear projection through shortcut connections is performed to match dimensions, resulting in the final output vector:

$$\mathbf{e}_{i'}^* = \mathbf{W}_s\boldsymbol{x} + \mathcal{F}\left(\boldsymbol{x}, \mathbf{W}_i\right), \tag{10}$$

where $\mathbf{W}_s$ is a linear transformation used to modify the number of channels to match the desired dimensions, $\mathcal{F}\left(\boldsymbol{x}, \mathbf{W}_i\right)$ denotes represents the residual mapping that will be learned. The output obtained from the residual network with shortcut connections is flattened into a one-dimensional vector. The vector representation is then concatenated with the item representation obtained through the previously described attention mechanism and fed into a fully connected layer for mapping into the embedding space, ultimately resulting in the embedding of the final item:

$$\mathbf{e}_i^* = \sigma\left(\mathbf{W}' \cdot [\mathbf{e}_i^*; \mathbf{e}_{i'}^*] + \mathbf{b}'\right), \tag{11}$$

where $[\mathbf{e}_i^*; \mathbf{e}_{i'}^*]$ denotes the concatenation of vectors along a certain dimension. $\mathbf{W}'$ denotes the weight matrix of the fully connected layer, $\mathbf{b}'$ denotes the bias term, and $\sigma(\cdot)$ denotes the rectified linear unit activation function.

Finally, we compute the matching score by taking the dot product of the user and item embedding, the formula is as follows:

$$\hat{y}_{(u,i)} = \mathbf{e}_u^{*\top}\mathbf{e}_i^*. \tag{12}$$

3.6. Training AMRN

We utilize a negative sampling strategy to calculate the loss. The loss is calculated by contrasting the similarity between positive samples and negative samples while applying a logistic sigmoid function, the formula is as follows:

$$\begin{aligned}\mathcal{L} = &\sum_{(h,r,t)\in\mathcal{E}} \sum_{(h,r,t')\notin\mathcal{E}} -\ln\sigma\left(\delta(h,r,t') - \delta(h,r,t')\right) + \\ &\sum_{(u,i)\in E^+,(u,j)\in E^-} -\ln\sigma\left(\mathbf{e}_u^{*\mathrm{T}} \cdot \mathbf{e}_i^* - \mathbf{e}_u^{*\mathrm{T}} \cdot \mathbf{e}_j^*\right) + \lambda\|\Theta\|_2^2,\end{aligned} \tag{13}$$

where $\sigma(\cdot)$ denotes the sigmoid function, Θ is the parameter set of our proposed model AMRN.

For clarity, the entire algorithm of AMRN is summarized in Algorithm 1.

Algorithm 1 AMRN

Input: User-item interaction graph $\mathcal{G}_{UI}$; knowledge graph $\mathcal{G}_{KG}$; relation path $p = (r_1, \dots, r_n)$; reliability score s; residual blocks number b; embedding dimension d.
Output: Score of user links to item.

```
for p ∈ P do
    for e_i ∈ E do
        Calculate resource amount R_p(i) via Eq. (1).
        Add invalid triplet into T^- via Eq. (2) when s exceeds set value.
    end for
end for
Calculate attention score δ(h, r, t) via Eq. (3).
Perform the softmax function via Eq. (4).
Aggregate e_h and e_{N_h} via Eq. (6).
High-order propagation in the l-th order neighborhood of the head entity h via Eq. (7).
Concatenate the representations to get e_u^* and e_i^* via Eq. (8).
Adopt residual mapping function via Eq. (9).
Linear projection by shortcut connections via Eq. (10).
Concatenate e_i^* and e_{i'}^*, perform fully connected layer via Eq. (11).
Calculate the matching score of the user and item via Eq. (12).
```

4. Experiments

4.1. Experimental settings

4.1.1. Datasets

The experiments are conducted on three real-world datasets, including Amazon-Book, Last-FM, and Yelp2018. The dataset statistics are presented in Table 2.

- **Amazon-Book**[1] comprises extensive user-book interactions from Amazon's book section [46]. It encompasses user reviews, star ratings, and various types of interactions with a vast collection of books available on the platform. This dataset is commonly used in research on recommender systems, personalized book recommendations, and sentiment analysis. Researchers and data scientists leverage this data to develop algorithms that can predict user preferences and assist in book discovery.
- **Last-FM**[2] is a rich source of user-music interaction data from the Last.fm music platform [22]. It includes user profiles, song listening histories, and detailed artist information. Researchers in the field of music recommender systems often utilize this dataset to study user behavior, create personalized music recommendations, and explore patterns in music consumption. It enables the

[1] https://jmcauley.ucsd.edu/data/amazon/
[2] https://grouplens.org/datasets/hetrec-2011/

Table 2
Statistics of the datasets.

		Amazon-Book	Last-FM	Yelp2018
User–Item Interaction	#Users	70,679	23,566	45,919
	#Items	24,915	48,123	45,538
	#Interactions	847,733	3,034,796	1,185,068
Knowledge Graph	#Entities	88,572	58,266	90,961
	#Relations	39	9	42
	#Triplets	2,557,746	464,567	1,853,704

development of models that enhance music discovery and user engagement.

- **Yelp2018**[3] is a comprehensive dataset containing user-generated reviews, ratings, and interactions with a diverse range of businesses and services listed on Yelp. These businesses span various categories, such as restaurants, shops, and entertainment venues. This dataset is invaluable for sentiment analysis, user opinion mining, and recommender systems. Researchers leverage it to gain insights into user preferences, evaluate business performance, and build recommendation models that assist users in discovering high-quality services and experiences.

4.1.2. Baselines

FM [12] (Factorization Machines) is a context-aware recommendation model based on matrix factorization techniques. It has linear complexity and performs well on sparse datasets for making predictions.

NFM [13] (Neural Factorization Machines) is a recommendation model designed for sparse data. It enhances the Factorization Machine (FM) by modeling non-linear and high-order feature interactions. NFM combines linear factorization machines with non-linear neural networks, resulting in strong predictive capabilities.

CKE [36] (Collaborative Knowledge base Embedding) integrates structured, textual, and visual content from a knowledge base. It uses a joint approach of knowledge base embedding and collaborative filtering to extract feature representations and capture implicit relationships between users and items.

ECFKG [47] (Embedding-based Collaborative Filtering with Knowledge Graph) is a personalized recommendation model leveraging KGs. It employs knowledge structures for fuzzy reasoning over KG embedding to generate personalized explanations.

BPRMF [48] (Bayesian Personalized Ranking Matrix Factorization) is a model used for ranking-based recommendations. It maximizes ranking loss to improve recommendation accuracy.

CFKG [37] (Collaborative Filtering with Knowledge Graph): CFKG integrates collaborative filtering with KGs to mitigate data sparsity and cold-start problems in recommendation tasks.

GC-MC [49] (Graph Convolutional Matrix Completion) employs graph convolutional networks for matrix completion tasks, filling missing values in user-item interaction matrices for recommendation.

RippleNet [25] (Ripples Propagating on the Networks) is a recommendation system model based on KGs and graph convolutional networks. It learns interaction patterns between users and items to improve recommendation quality.

KGAT [22] (Knowledge Graph Attention Network) is a KG-enhanced recommendation model that uses attention mechanisms to consider information from the KG for improved personalization.

JNSKR [41] (Jointly Non-Sampling learning model for Knowledge graph enhanced Recommendation) model combines social influence and KG information to enhance recommendation system performance.

KR-GCN [40] (Knowledge-aware Reasoning with Graph Convolutional Networks) is a KG-aware graph convolutional network used for recommendation system tasks, providing a better understanding of user-item relationships.

[3] https://www.yelp.com/dataset/challenge

Table 3
Experimental results on three datasets.

Methods	Amazon-Book		Last-FM		Yelp2018	
	Recall	NDCG	Recall	NDCG	Recall	NDCG
FM	0.1345	0.0886	0.0778	0.1181	0.0627	0.0768
NFM	0.1366	0.0913	0.0829	0.1214	0.0660	0.0810
CKE	0.1343	0.0885	0.0736	0.1184	0.0657	0.0805
ECFKG	0.1142	0.0770	0.0723	0.1143	0.0522	0.0644
BPRMF	0.1336	0.0694	0.0727	0.1147	0.0644	0.0767
CFKG	0.1142	0.0770	0.0723	0.1143	0.5220	0.0644
GC-MC	0.1316	0.0874	0.0818	0.1253	0.0659	0.0790
RippleNet	0.1336	0.0910	0.0791	0.1238	0.0664	0.0822
KGAT	0.1489	0.1006	0.0870	0.1325	0.0712	0.0867
JNSKR	0.1558	0.1068	0.0812	0.1306	0.0749	0.0917
KR-GCN	0.1573	0.1076	0.0892	0.1353	0.0758	0.0921
AMRN (ours)	**0.1588**	**0.1077**	**0.0902**	**0.1380**	**0.0763**	**0.0925**

Note: The best results are in bold.

4.1.3. Evaluation metrics

In information retrieval and recommender systems, Recall@k is a metric that assesses a model's ability to capture relevant items within its top-k predictions. It quantifies the fraction of true relevant items correctly included within the top-k recommendations. The formula for Recall@k is:

$$\text{Recall@}k = \frac{\text{TP@}k}{\text{TP@}k + \text{FN@}k}, \tag{14}$$

where TP@k represents the number of correctly predicted relevant items among the top-k recommendations generated by the model, FN@k counts the relevant items that were not included within the top-k recommendations.

NDCG@k is an evaluation metric that assesses the quality of a ranked list or recommendation system up to a specified position, k, as follows:

$$\text{NDCG@}k = \frac{\text{DCG@}k}{\text{IDCG@}k}, \tag{15}$$

where

$$\text{DCG@}k = \sum_{i=1}^{k} \frac{rel_i}{\log_2(i+1)} \tag{16}$$

and

$$\text{IDCG@}k = \sum_{i=1}^{k} \frac{2^{rel_i} - 1}{\log_2(i+1)}. \tag{17}$$

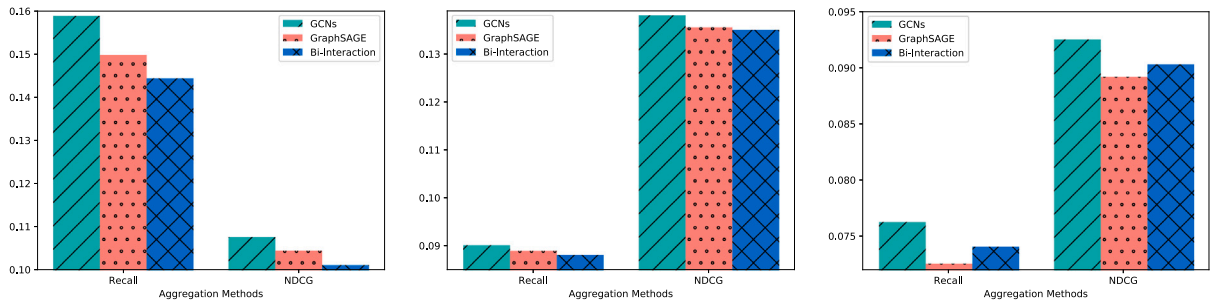
In Eqs. (16) and (17), rel_i represents the relevance score assigned to the item in position i within the ranked list. Typically, this score is binary, signifying the item's relevance (1) or lack thereof (0).

4.1.4. Parameter setting

In our implementation, the parameter settings for each baseline are configured with their respective optimal values to ensure fair comparisons. For our proposed model AMRN, we employ the Xavier initializer and Adam optimizer to initialize and optimize the model parameters, respectively. The model is trained for 500 epochs, and the batch size is set as 1024. The learning rate and dropout rate are set as 0.0001 and 0.1, respectively. To reduce computational cost, we limit the length of paths for resource propagation to 3 steps and consider paths with reliability score larger than 0.01. We set the embedding dimensions d as 64 for three datasets. For the number of residual blocks b, we set $b = 2$ for the datasets Amazon-Book and Last-FM, and $b = 3$ for the dataset Yelp2018. A detailed analysis of the parameters d and b is reported in Section 4.3.

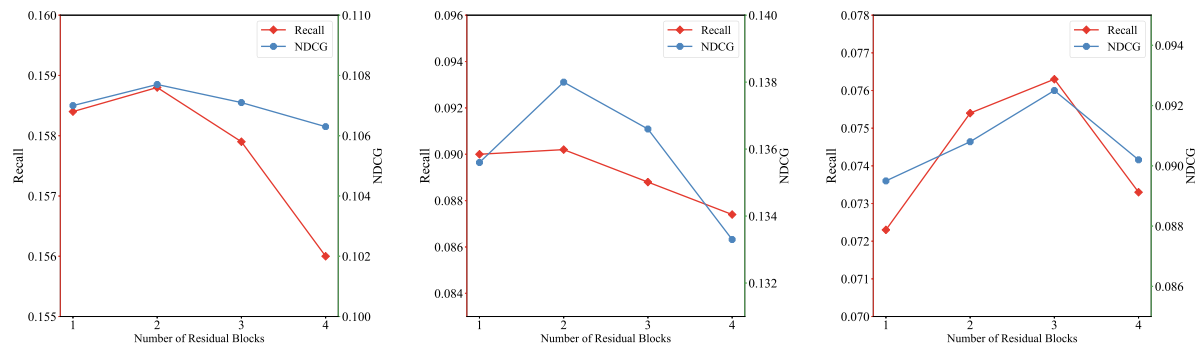
4.2. Comparison results

The results of the comparison, presented in Table 3, showcase the performance of various baseline methods in terms of Recall@k and NDCG@k (where $k = 20$ for each metric) on the three datasets.



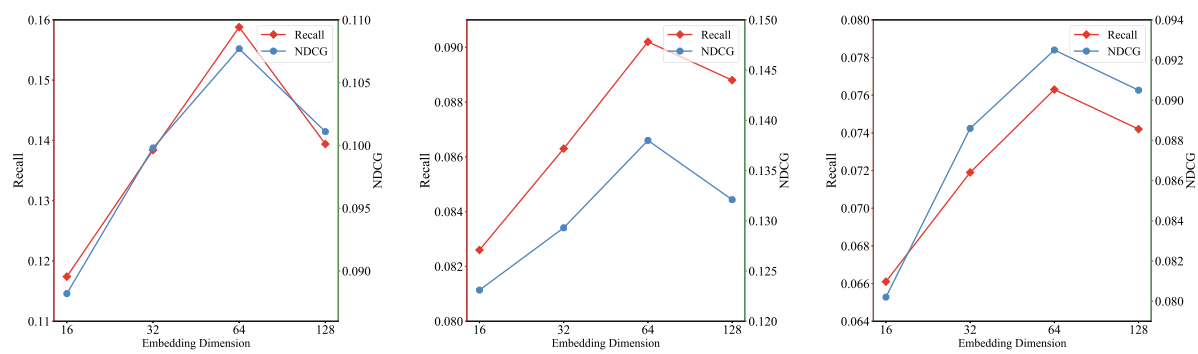
(a) Amazon-Book (k=20) (b) Last-FM (k=20) (c) Yelp2018 (k=20)

Fig. 3. Impact of aggregation methods on the three datasets.



(a) Amazon-Book (k=20) (b) Last-FM (k=20) (c) Yelp2018 (k=20)

Fig. 4. Impact of the number of residual blocks on the three datasets.



(a) Amazon-Book (k=20) (b) Last-FM (k=20) (c) Yelp2018 (k=20)

Fig. 5. Impact of embedding dimension on the three datasets.

In comparison to all baseline methods, AMRN demonstrates superior performance across two evaluation metrics on the three datasets. Specifically, on the Last-FM dataset, AMRN achieves significant improvements of 1.121% in Recall@k and 1.995% in NDCG@k. On the other two datasets, AMRN shows modest improvements, with a 0.954% boost in Recall@20 and a 0.0923% enhancement in NDCG@20 for Amazon-Book and a 0.660% improvement in Recall@20 and a 0.434% enhancement in NDCG@20 for Yelp2018. Compared with FM, NFM, and BPRMF, AMRN places greater emphasis on the information representation of users and entities during the embedding process. In contrast to CKE and ECFKG, AMRN harnesses higher-order information transmitted by entities within its neighborhood. While GC-MC and RippleNet capture higher-order connections, they fail to effectively identify the significance of edges between entities. As an extension of KGAT, AMRN employs a more refined deep residual network mechanism for KG enhancement, enabling a more profound exploration of the structural and semantic information of entities in the KG. Compared with models such as JNSKR and KR-GCN that also leverage KGs to enhance recommendations, AMRN utilizes deep CNNs to combine features. It introduces shortcut connections to achieve residual mapping, alleviating the issues of gradient explosion and gradient vanishing.

Three conclusions can be drawn based on the superior performance of the AMRN model. First, the resource allocation algorithm employed during path extraction effectively filters out unreliable paths within the KG, thus providing items with more comprehensive and reliable enhanced representations. The reason is intuitive that the semantic composition of relations can be leveraged to measure path reliability, optimizing the enhanced representation from the KG. Second, AMRN leverages an attention mechanism to assign varying degrees of importance to different entities and relations in the KG, effectively capturing higher-order connectivity and long-range dependencies. Subsequently, GCN is employed to aggregate representation information of users and items from all propagation layers, achieving information aggregation. Third, we employ a residual network with shortcut connections to

Table 4
Performance comparison of ablation study.

Datasets	Metrics	w/o P&R	w/o P	w/o R	w/o A(i)	AMRN
Amazon-Book	Recall	0.1378	0.1527	0.1499	0.1506	**0.1588**
	NDCG	0.0932	0.1042	0.1013	0.1022	**0.1077**
Last-FM	Recall	0.0826	0.0897	0.0881	0.0890	**0.0902**
	NDCG	0.1288	0.1373	0.1333	0.1362	**0.1380**
Yelp2018	Recall	0.0691	0.0726	0.0756	0.0728	**0.0763**
	NDCG	0.0844	0.0880	0.0893	0.0891	**0.0925**

Note: The best results are in bold.

delve deep into the feature information associated with items in the KG. In the residual block, the mainstem utilizes CNNs for feature learning. Meanwhile, shortcut mechanisms are introduced to facilitate skip connections, effectively alleviating the challenges associated with gradient explosion and convergence issues resulting from excessively deep networks.

4.3. Study of AMRN

To investigate the impact of different components on AMRN's performance, we conduct comprehensive experiments to analyze the influence of aggregation method, residual block, and embedded dimension on three datasets, respectively.

4.3.1. Impact of aggregation method

The attention embedding propagation phase is a crucial component where different aggregation methods have varying effects on the performance of the model. To investigate the impact of using different aggregation methods during this phase, we conducted experiments using three aggregation methods: GCNs (Graph Convolutional Networks), GraphSAGE, and Bi-Interaction on the three datasets. As illustrated in Fig. 3, the results clearly validate that employing GCNs as the aggregation method yields the best performance on all the datasets. Compared with the other two aggregation methods, GCNs are often more computationally efficient, especially when dealing with large-scale graphs. This efficiency is due to their ability to aggregate information from neighboring entities in a more localized and parallelizable manner. Additionally, GCNs inherently consider the global structure of the graph during message passing, enabling them to capture more complex relationships and interactions among entities.

4.3.2. Impact of residual block

Although increasing the number of network layers allows the combination of higher-level abstract features, an excessive number of layers can introduce noise into the resulting feature representations. A residual network based on shortcut connections is a residual block, and the number of residual blocks determines the depth of a network. To investigate the impact of the number of residual blocks on the performance of AMRN, experiments are conducted using varying numbers $\{1, 2, 3, 4\}$ of residual blocks on each dataset. As shown in Fig. 4, the results demonstrate a trend of performance improvement followed by degradation with increasing model depth. The evaluation metrics are the highest when the number of residual blocks is set to 2 or 3. However, further increasing the number of residual blocks beyond this range leads to a decrease in performance. This indicates that increasing depth within a suitable range can enhance recommendation performance, but excessive depth can deteriorate the performance.

4.3.3. Impact of embedded dimension

To investigate the impact of embedding dimension on the performance of our model, we conducted experiments on the three datasets with embedding dimensions ranging from $\{d = 16, 32, 64, 128\}$. As shown in Fig. 5, with an increase in embedding dimensionality, Recall @20 and NDCG@20 initially exhibit an upward trend, reaching their peak performance when the embedding dimension is set to 64 for AMRN. However, as the embedding dimension becomes excessively large, performance starts to deteriorate. This is because an overly large embedding dimension can lead to several drawbacks, including increased model complexity, higher computational overhead, and susceptibility to overfitting. An appropriate embedding dimension can strike a balance between model expressiveness and computational efficiency.

4.4. Ablation study

To investigate the impact of path extraction (denoted as P), attention propagation (denoted as A), and residual network learning (denoted as R) on the performance of the model, ablation experiments are conducted on the three datasets. These experiments include w/o P&R, w/o P, w/o R, and w/o A(i), where w/o represents without. For example, w/o A(i) signifies that the attention mechanism was not employed to learn item representations.

The experimental results presented in Table 4 (with evaluation metrics at k = 20) demonstrate that the complete model AMRN exhibits the best performance, followed by w/o P and w/o A(i). It can be observed that attention mechanisms effectively capture user-item interactions, along with higher-order connections and long-range dependencies. It is evident that the performance of w/o R is better than w/o P&R, indicating that filtering out unreliable paths through computational resource amount can provide a higher-quality graph structure for subsequent representation learning. The comparison of performances of w/o P and w/o P&R shows that the residual network based on the shortcut connections applies deep CNNs to capture features of the KG, enhancing the representation of item features using the KG. The performance of w/o P is better than that of w/o R, indicating that the residual network module plays a more significant role than the path extraction module. This suggests that the use of the KG allows better capturing of item features, leading to richer and more comprehensive representations, thereby enhancing the recommendation effectiveness.

5. Conclusion

In this article, we propose AMRN, a recommender system that utilizes an attention mechanism and deep residual networks with shortcut connections. AMRN is designed to address challenges commonly encountered in traditional KG-based recommender methods, including gradient instability in deep networks and the presence of unreliable paths within the KG. First, we employ a resource allocation algorithm to evaluate the path reliability in the KG, allowing us to filter out unreliable paths effectively. Following path extraction, we utilize an attention mechanism to capture semantic correlations and structural information of the KG. Then, we design a deep residual network with shortcut connections, enabling the model to effectively fuse advanced and abstract features. This architecture mitigates issues like gradient instability, which can arise with excessive network depth. Our experiments on standard datasets demonstrate that the proposed KG-enhanced recommender system outperforms traditional methods. In our future work, further research aims to further investigate techniques for incorporating more dynamic and up-to-date knowledge, as well as exploring methods for handling noisy or incomplete data within the graph. Additionally, research can explore enhancements to the attention mechanism, residual network, and path extraction techniques to achieve even better results.

CRediT authorship contribution statement

Weisheng Li: Writing – original draft, Methodology, Conceptualization. **Hao Zhong:** Visualization, Investigation. **Junming Zhou:** Validation, Software. **Chao Chang:** Supervision, Conceptualization. **Ronghua Lin:** Writing – review & editing, Visualization. **Yong Tang:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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